GridSpike: Byzantine-robust federated learning for edge event classification in IEC 61850 substations

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Abstract

Distribution substations instrumented to IEC 61850 produce streams classifiable at the edge, but utilities cannot pool the data and the edge is energy-constrained and adversarially exposed. We present GridSpike, a federated learning framework and an open benchmark generator: a quasi-static OpenDSS simulation of five feeders with synthetic event overlays, non-IID by feeder identity and Dirichlet label skew. Three classifiers—convolutional–recurrent, spiking and selective state-space (Mamba)—are learned over ten substations behind a three-stage Byzantine-admission pipeline: scale-free fidelity scoring, simulated PBFT agreement and robust aggregation, hash-chained per round. Across an expanded battery of 6,830 run records the models trace an accuracy–energy–latency trade-off. Under a leakage-controlled chronological split the two dense models tie at the top (macro-F_1 0.596 reference, 0.590 state-space; paired $p = 0.56$), and the spiking model—at 0.7% of the reference's per-inference energy—collapses to 0.13; the higher 0.77/0.60/0.16 is a random-split ceiling inflated by window leakage, and the state-space lead it shows resolves to the tie. The reference's p99 inference (3.3 ms on GPU) would consume 82% of a co-resident 4 ms GOOSE fast-message window; the other two exceed it. Under a defense-aware adversary the pipeline beats neither plain averaging nor robust aggregation alone: it buys auditability, not accuracy; usable local differential privacy is incompatible with admission. Sweeping that adversary's in-cone magnitude 10× breaks plain averaging, not the geometric median; at the endpoints honest rejection falls from 0.80 to 0.47 while attacker admission does not fall. Batteries regenerate from the committed manifests; per-run configuration hashes verify canonically for 67% of the original battery's records.

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Highlights

- Two dense model families tie on accuracy inside a three-way edge trade-off
- A quasi-static OpenDSS generator makes the federation non-IID by construction
- A high-magnitude in-cone attacker breaks plain averaging, not the geometric median
- Under a defense-aware attacker, agreement buys auditability, not model accuracy
- Usable local differential privacy is incompatible with the Byzantine admission gate

CRediT authorship contribution statement

The CRediT taxonomy defines fourteen roles; list each author once, with every role that applies, and delete the roles that do not. The corresponding author is responsible for ensuring that this statement is accurate and agreed by all authors.

[?? author 1 name]: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition.

[?? author 2 name]: [?? CRediT roles for this author].

[?? author n name]: [?? CRediT roles for this author].

Declaration of competing interest

The authors declare the following financial interests / personal relationships which may be considered as potential competing interests: [?? competing interests, or "none"].

If there are none, the statement reads: *The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.*

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Data and code availability

All data behind the primary results are generated by the released physics-based simulation. One third-party dataset is used, for the real-trace sanity check only: the publicly available MSU/ORNL power-system attack and disturbance dataset, used unmodified under its published terms. The complete framework – simulation, models, federated engine, attack suite, consensus pipeline, experiment runner, analysis code, tests and live console – is released open source, with an archived release carrying a persistent identifier. During review the artifact is provided through an anonymized link; the de-anonymized repository and archive identifiers are [?? repository URL] and [?? archive DOI]. The software package retains its development name `fedgrid` in the repository and import paths; the system is named GridSpike in the manuscript.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author(s) used [?? name of tool / service] in order to [?? reason: e.g. draft and edit sections of the manuscript]. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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